

```
In [2]: #!/pip install matplotlib
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [3]: df=pd.read_csv("Social_Network_Ads.csv")
df.head()
```

```
Out[3]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [4]: df.dtypes
```

```
Out[4]: User ID          int64
Gender          object
Age            int64
EstimatedSalary int64
Purchased       int64
dtype: object
```

```
In [5]: (df.isnull()).sum()
```

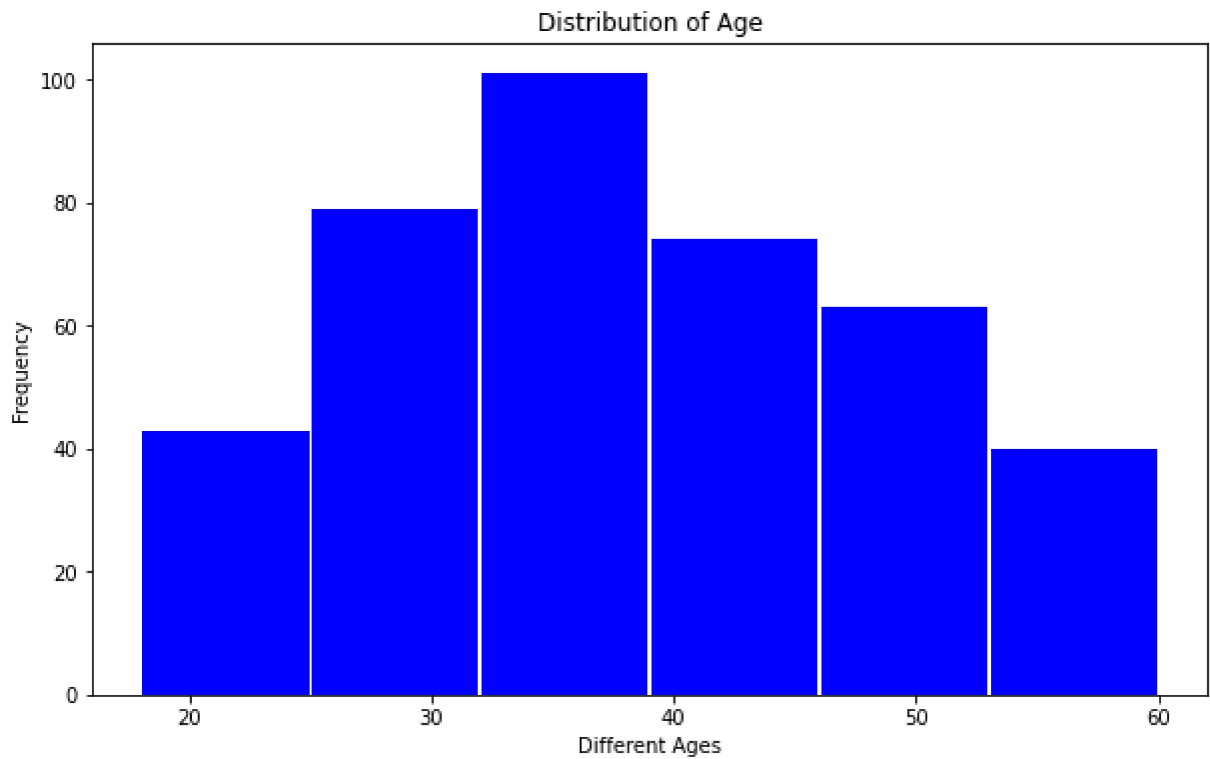
```
Out[5]: User ID          0
Gender          0
Age            0
EstimatedSalary 0
Purchased       0
dtype: int64
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   User ID         400 non-null   int64
1   Gender          400 non-null   object
2   Age             400 non-null   int64
3   EstimatedSalary 400 non-null   int64
4   Purchased       400 non-null   int64
dtypes: int64(4), object(1)
memory usage: 15.8+ KB
```

```
In [8]: plt.figure(figsize = (10,6))
plt.hist(df['Age'], bins = 6, color = 'blue', rwidth = 0.98)
plt.title('Distribution of Age')
plt.xlabel('Different Ages')
plt.ylabel('Frequency')
```

```
Out[8]: Text(0, 0.5, 'Frequency')
```



```
In [9]: X = df.iloc[:,[2,3]].values
print(X)
```

```
[[ 19 19000]
 [ 35 20000]
 [ 26 43000]
 [ 27 57000]
 [ 19 76000]
 [ 27 58000]
 [ 27 84000]
 [ 32 150000]
 [ 25 33000]
 [ 35 65000]
 [ 26 80000]
 [ 26 52000]
 [ 20 86000]
 [ 32 18000]
 [ 18 82000]
 [ 29 80000]
 [ 47 25000]
 [ 45 26000]
 [ 46 28000]
 [ 48 29000]
 [ 45 22000]
 [ 47 49000]
 [ 48 41000]
 [ 45 22000]
 [ 46 23000]
 [ 47 20000]
 [ 49 28000]
 [ 47 30000]
 [ 29 43000]
 [ 31 18000]
 [ 31 74000]
 [ 27 137000]
 [ 21 16000]
 [ 28 44000]
 [ 27 90000]
 [ 35 27000]
 [ 33 28000]
 [ 30 49000]
 [ 26 72000]
 [ 27 31000]
 [ 27 17000]
 [ 33 51000]
 [ 35 108000]
 [ 30 15000]
 [ 28 84000]
 [ 23 20000]
 [ 25 79000]
 [ 27 54000]
 [ 30 135000]
 [ 31 89000]
 [ 24 32000]
 [ 18 44000]
 [ 29 83000]
 [ 35 23000]
 [ 27 58000]
 [ 24 55000]
```

```
[ 23 48000]
[ 28 79000]
[ 22 18000]
[ 32 117000]
[ 27 20000]
[ 25 87000]
[ 23 66000]
[ 32 120000]
[ 59 83000]
[ 24 58000]
[ 24 19000]
[ 23 82000]
[ 22 63000]
[ 31 68000]
[ 25 80000]
[ 24 27000]
[ 20 23000]
[ 33 113000]
[ 32 18000]
[ 34 112000]
[ 18 52000]
[ 22 27000]
[ 28 87000]
[ 26 17000]
[ 30 80000]
[ 39 42000]
[ 20 49000]
[ 35 88000]
[ 30 62000]
[ 31 118000]
[ 24 55000]
[ 28 85000]
[ 26 81000]
[ 35 50000]
[ 22 81000]
[ 30 116000]
[ 26 15000]
[ 29 28000]
[ 29 83000]
[ 35 44000]
[ 35 25000]
[ 28 123000]
[ 35 73000]
[ 28 37000]
[ 27 88000]
[ 28 59000]
[ 32 86000]
[ 33 149000]
[ 19 21000]
[ 21 72000]
[ 26 35000]
[ 27 89000]
[ 26 86000]
[ 38 80000]
[ 39 71000]
[ 37 71000]
```

```
[ 38 61000]
[ 37 55000]
[ 42 80000]
[ 40 57000]
[ 35 75000]
[ 36 52000]
[ 40 59000]
[ 41 59000]
[ 36 75000]
[ 37 72000]
[ 40 75000]
[ 35 53000]
[ 41 51000]
[ 39 61000]
[ 42 65000]
[ 26 32000]
[ 30 17000]
[ 26 84000]
[ 31 58000]
[ 33 31000]
[ 30 87000]
[ 21 68000]
[ 28 55000]
[ 23 63000]
[ 20 82000]
[ 30 107000]
[ 28 59000]
[ 19 25000]
[ 19 85000]
[ 18 68000]
[ 35 59000]
[ 30 89000]
[ 34 25000]
[ 24 89000]
[ 27 96000]
[ 41 30000]
[ 29 61000]
[ 20 74000]
[ 26 15000]
[ 41 45000]
[ 31 76000]
[ 36 50000]
[ 40 47000]
[ 31 15000]
[ 46 59000]
[ 29 75000]
[ 26 30000]
[ 32 135000]
[ 32 100000]
[ 25 90000]
[ 37 33000]
[ 35 38000]
[ 33 69000]
[ 18 86000]
[ 22 55000]
[ 35 71000]
```

```
[ 29 148000]
[ 29 47000]
[ 21 88000]
[ 34 115000]
[ 26 118000]
[ 34 43000]
[ 34 72000]
[ 23 28000]
[ 35 47000]
[ 25 22000]
[ 24 23000]
[ 31 34000]
[ 26 16000]
[ 31 71000]
[ 32 117000]
[ 33 43000]
[ 33 60000]
[ 31 66000]
[ 20 82000]
[ 33 41000]
[ 35 72000]
[ 28 32000]
[ 24 84000]
[ 19 26000]
[ 29 43000]
[ 19 70000]
[ 28 89000]
[ 34 43000]
[ 30 79000]
[ 20 36000]
[ 26 80000]
[ 35 22000]
[ 35 39000]
[ 49 74000]
[ 39 134000]
[ 41 71000]
[ 58 101000]
[ 47 47000]
[ 55 130000]
[ 52 114000]
[ 40 142000]
[ 46 22000]
[ 48 96000]
[ 52 150000]
[ 59 42000]
[ 35 58000]
[ 47 43000]
[ 60 108000]
[ 49 65000]
[ 40 78000]
[ 46 96000]
[ 59 143000]
[ 41 80000]
[ 35 91000]
[ 37 144000]
[ 60 102000]
```

```
[ 35 60000]
[ 37 53000]
[ 36 126000]
[ 56 133000]
[ 40 72000]
[ 42 80000]
[ 35 147000]
[ 39 42000]
[ 40 107000]
[ 49 86000]
[ 38 112000]
[ 46 79000]
[ 40 57000]
[ 37 80000]
[ 46 82000]
[ 53 143000]
[ 42 149000]
[ 38 59000]
[ 50 88000]
[ 56 104000]
[ 41 72000]
[ 51 146000]
[ 35 50000]
[ 57 122000]
[ 41 52000]
[ 35 97000]
[ 44 39000]
[ 37 52000]
[ 48 134000]
[ 37 146000]
[ 50 44000]
[ 52 90000]
[ 41 72000]
[ 40 57000]
[ 58 95000]
[ 45 131000]
[ 35 77000]
[ 36 144000]
[ 55 125000]
[ 35 72000]
[ 48 90000]
[ 42 108000]
[ 40 75000]
[ 37 74000]
[ 47 144000]
[ 40 61000]
[ 43 133000]
[ 59 76000]
[ 60 42000]
[ 39 106000]
[ 57 26000]
[ 57 74000]
[ 38 71000]
[ 49 88000]
[ 52 38000]
[ 50 36000]
```

```
[ 59 88000]
[ 35 61000]
[ 37 70000]
[ 52 21000]
[ 48 141000]
[ 37 93000]
[ 37 62000]
[ 48 138000]
[ 41 79000]
[ 37 78000]
[ 39 134000]
[ 49 89000]
[ 55 39000]
[ 37 77000]
[ 35 57000]
[ 36 63000]
[ 42 73000]
[ 43 112000]
[ 45 79000]
[ 46 117000]
[ 58 38000]
[ 48 74000]
[ 37 137000]
[ 37 79000]
[ 40 60000]
[ 42 54000]
[ 51 134000]
[ 47 113000]
[ 36 125000]
[ 38 50000]
[ 42 70000]
[ 39 96000]
[ 38 50000]
[ 49 141000]
[ 39 79000]
[ 39 75000]
[ 54 104000]
[ 35 55000]
[ 45 32000]
[ 36 60000]
[ 52 138000]
[ 53 82000]
[ 41 52000]
[ 48 30000]
[ 48 131000]
[ 41 60000]
[ 41 72000]
[ 42 75000]
[ 36 118000]
[ 47 107000]
[ 38 51000]
[ 48 119000]
[ 42 65000]
[ 40 65000]
[ 57 60000]
[ 36 54000]
```



```
[ 58 144000]
[ 35 79000]
[ 38 55000]
[ 39 122000]
[ 53 104000]
[ 35 75000]
[ 38 65000]
[ 47 51000]
[ 47 105000]
[ 41 63000]
[ 53 72000]
[ 54 108000]
[ 39 77000]
[ 38 61000]
[ 38 113000]
[ 37 75000]
[ 42 90000]
[ 37 57000]
[ 36 99000]
[ 60 34000]
[ 54 70000]
[ 41 72000]
[ 40 71000]
[ 42 54000]
[ 43 129000]
[ 53 34000]
[ 47 50000]
[ 42 79000]
[ 42 104000]
[ 59 29000]
[ 58 47000]
[ 46 88000]
[ 38 71000]
[ 54 26000]
[ 60 46000]
[ 60 83000]
[ 39 73000]
[ 59 130000]
[ 37 80000]
[ 46 32000]
[ 46 74000]
[ 42 53000]
[ 41 87000]
[ 58 23000]
[ 42 64000]
[ 48 33000]
[ 44 139000]
[ 49 28000]
[ 57 33000]
[ 56 60000]
[ 49 39000]
[ 39 71000]
[ 47 34000]
[ 48 35000]
[ 48 33000]
[ 47 23000]
```

```
[ 45 45000]
[ 60 42000]
[ 39 59000]
[ 46 41000]
[ 51 23000]
[ 50 20000]
[ 36 33000]
[ 49 36000]]
```

```
In [10]: y = df.iloc[:,4].values
         print(y)
```

```
[0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 0 0 0 1 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 0
 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 1 0 1 0 1 1 0 0 0 1 0 0 0 1 0 1
 1 1 0 0 1 1 0 1 1 0 1 1 0 1 0 0 0 1 1 0 1 1 0 1 0 1 0 1 0 1 0 0 1 1 0 1 0 0 1
 1 0 1 1 0 1 1 0 0 1 0 0 1 1 1 1 0 1 1 1 1 0 1 1 0 1 0 1 0 1 1 1 1 0 0 0
 1 1 0 1 1 1 1 1 0 0 0 1 1 0 0 1 0 1 0 1 1 0 1 0 1 1 0 1 1 0 0 0 1 1 0 1 0
 0 1 0 1 0 0 1 1 0 0 1 1 0 1 1 0 0 1 0 1 0 1 1 1 0 1 1 1 0 1 1 1 0 1 1 0 1
 1 1 0 1 0 1 0 0 1 1 0 1 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 0 1 1 1 0 1]
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```